SOLUTION DESIGN DOCUMENT (SDD)

RPA solution • Supply Chain Material Tracker & Dashboarding

1. Design purpose

This SDD outlines a modular Automation Anywhere solution for collecting supply-chain material data from SAP dumps, web portals and OneDrive, consolidating and validating the data, saving the reporting output, and supporting Power BI refresh. Confirm source-specific details and enterprise controls before implementation.

2. Solution architecture

Component	Design
Trigger / scheduler	Approved daily schedule or manual trigger with configurable run parameters.
Automation Anywhere	Controller task, reusable extraction tasks, transformation/validation steps and exception handling.
Source adapters	SAP dump file reader, web portal interaction/download and OneDrive file access.
Staging / transformation	Controlled working folder and Excel/CSV processing using the approved common schema.
Output repository	Approved OneDrive/shared folder with restricted permissions and versioned output.
Power BI	Dataset/dashboard refresh through the report owner's approved refresh mechanism.
Logging / notifications	Run summary, source-level status, data-quality metrics and sanitized errors.

3. High-level workflow

1. Initialize run ID, date, configuration and output paths.
2. Validate configuration and access to each source.
3. Extract or download data from SAP dumps, web portals and OneDrive.
4. Validate source files and map fields to the canonical schema.

5. Normalize values, apply approved status mappings and deduplicate.

6. Run data-quality checks and produce a reconciliation summary.

7. Write a new output version to the approved repository.

8. Trigger or prepare the Power BI refresh and verify its outcome where supported.

9. Publish run status and exceptions, then clean temporary files safely.

4. Suggested bot modules

Module / task bot	Responsibility
Main_Controller	Orchestrates modules, run state, cleanup and final status.
Load_Configuration	Reads source list, paths, expected columns, criticality and run parameters.
Extract_SAP_Dumps	Locates and reads approved SAP dump files; records source timestamp and counts.
Extract_Web_Portals	Authenticates through approved methods, navigates to configured pages and downloads data.
Extract_OneDrive	Retrieves files from approved folders and records file metadata.
Validate_Source_Data	Checks schema, mandatory fields, formats, freshness and file integrity.
Transform_And_Consolidate	Maps fields, standardizes values, combines sources and applies approved duplicate rules.
Data_Quality_Checks	Calculates rejected-row counts, duplicate counts, missing fields and reconciliation results.
Publish_Output	Saves validated output using versioned naming and safe replacement behavior.
Refresh_PowerBI	Invokes the approved refresh method and records the returned status where available.
Write_Run_Log_Notify	Stores sanitized diagnostics and sends success/exception notifications.

5. Configuration and data model

Parameter / field	Design rule	Validation
Run_ID	Unique identifier for each	Required in run log and output

	execution	metadata.
Run_Date / cutoff	Business-defined date and late-source cutoff	Must follow approved timezone/calendar rules.
Source_ID	Configured identifier for each source	Must be in the approved source list.
Material_ID	Canonical material identifier	Trim/normalize only as approved; preserve leading zeros if meaningful.
Material_Status	Canonical status value	Map only through approved source-to-canonical mappings.
Source_Timestamp	Time data was extracted or last updated	Retain to support freshness checks.
Required business fields	Defined in the data dictionary	Missing values trigger reject/flag rules.
Output_Path / File_Name	Approved repository and naming pattern	Verify write permission and prevent accidental overwrite.

6. Source extraction design

- SAP dumps: use configured folder/file patterns and verify the file date, format and expected columns before processing.
- Web portals: use stable selectors or supported browser automation, explicit waits and checks for successful download; avoid coordinate-only automation where possible.
- OneDrive: access only approved folders and use the organization's authorized authentication mechanism.
- Record source name, extraction timestamp, file name, source record count and success/failure for each source.
- Use bounded retries for transient connectivity issues; do not repeatedly download or process stale data without a freshness check.

7. Transformation and reconciliation

- Map source columns to a documented canonical schema and preserve traceability to the originating source.
- Normalize whitespace, approved date formats and status labels without changing business meaning.

- Use an approved composite key for deduplication; where source records conflict, apply an agreed precedence rule or flag for review.
- Compare input, accepted and rejected record counts and record the reconciliation totals.
- Keep raw inputs read-only and separate from transformed outputs to support audit and troubleshooting.
- Do not silently fill missing material statuses or infer inventory availability.

8. Output and Power BI integration

- Write the consolidated dataset to a temporary/new file first, validate it, then publish it using the approved versioning strategy.
- Use consistent column names, data types, encoding and date formats compatible with the Power BI model.
- Refresh Power BI through the approved gateway/API/scheduled-refresh mechanism confirmed by the report owner; the precise mechanism is not specified in the source brief.
- Where refresh status is available, record request time, completion state and failure details.
- Show the last successful refresh timestamp or a clear stale-data warning in the operational status process.

9. Error handling and recovery

Error category	Handling strategy	Retry / recovery
Invalid configuration	Fail fast and identify the missing parameter.	Correct configuration; rerun.
Source timeout / temporary failure	Log source-specific error and preserve successful extractions.	Bounded retry, then escalate.
Missing critical source	Apply the approved full-stop or partial-publication policy.	Wait for source or obtain process-owner approval.
Schema drift	Quarantine affected source and stop its transformation.	Update mapping after owner approval and regression test.
Bad rows / conflicting statuses	Reject or flag according to the data-quality rule.	Business owner reviews exceptions.
Publish failure	Do not replace the last known-good output.	Check permissions/storage and rerun publish safely.
Power BI refresh failure	Keep dataset status separate from bot extraction status.	Alert report owner; retry only within policy.

10. Security, audit and operations

- Use least-privilege service accounts and approved credential storage; do not embed secrets in bot code.
- Restrict source files, consolidated reports and logs to authorized users.
- Avoid logging credentials or unnecessary sensitive supplier/material information.
- Record bot version, run ID, source status, record counts, output version and sanitized error details.
- Apply approved retention and cleanup rules to temporary downloads and staging files.
- Separate development, test and production configuration and use change approval before deployment.

11. Test strategy

Test case	Expected result
All sources available with valid data	Consolidated output is created and quality checks pass.
One non-critical source unavailable	Approved partial-data behavior is applied and clearly logged.
Critical source missing	Bot follows the full-stop/approval policy; no misleading final output.
Unexpected column name or file format	Source is flagged or quarantined; no guessed mapping.
Duplicate material records	Approved key and precedence rules are applied consistently.
Missing material ID or status	Rows are rejected/flagged according to business rules.
OneDrive output permission denied	Publish fails safely and last known-good output remains intact.
Power BI refresh fails	Extraction and refresh outcomes are separately logged; owner is notified.
Stale source file	Freshness check flags the file and follows the approved cutoff policy.

12. Deployment and support

- Build and validate source adapters in a non-production environment using approved test data.

- Complete reconciliation tests against manually prepared sample outputs.
- Confirm scheduler, runner availability, source credentials, storage permissions and Power BI refresh configuration.
- Obtain sign-off from supply chain, source owners, reporting owner and RPA support before production release.
- Document rollback: disable the schedule, restore the prior bot version and use the approved manual reporting process.
- Monitor initial production runs for source availability, data quality, runtime and dashboard freshness.

13. Items to confirm before implementation

- Exact SAP dump names/formats and web portal URLs or navigation paths.
- OneDrive access method, folder structure and expected file naming.
- Canonical data dictionary, material unique key, duplicate precedence and status mapping.
- Critical vs optional sources and policy for partial reporting.
- Power BI dataset/model name, refresh mechanism, gateway dependencies and access controls.
- Run schedule, cutoff time, alert recipients, log retention and support escalation.